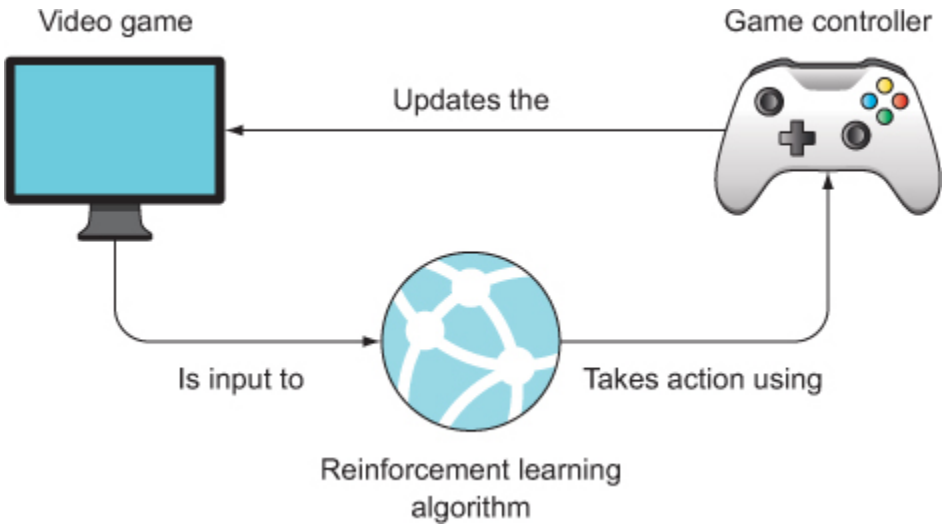
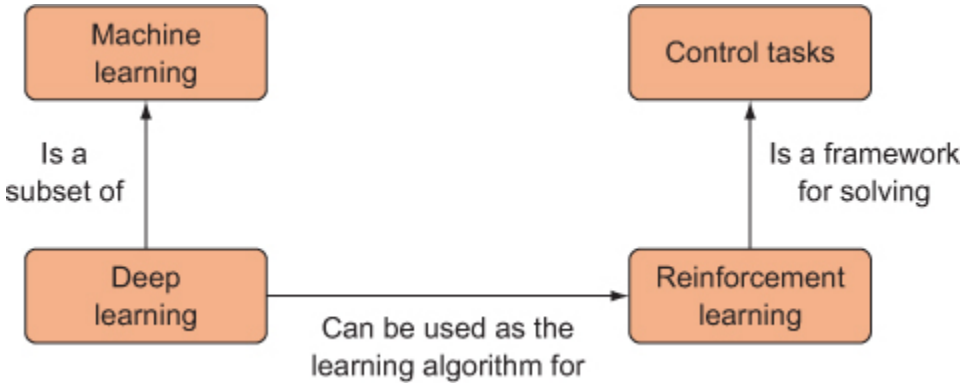


Figure 1.5. Deep learning is a subfield of machine learning. Deep learning algorithms can be used to power RL approaches to solving control tasks.



One added complexity of moving from image processing to the domain of control tasks is the additional element of time. With image processing, we usually train a deep learning algorithm on a fixed data set of images. After a sufficient amount of training, we typically get a high-performance algorithm that we can deploy to some new, unseen images. We can think of the data set as a "space" of data, where similar images are closer together in this abstract space and distinct images are farther apart (figure 1.6).

Figure 1.6. This graphical depiction of words in a 2D space shows each word as a colored point. Similar words cluster together, and dissimilar words are farther apart. Data naturally lives in some kind of "space" with similar data living closer together. The labels A, B, C, and D point to particular clusters of words that share some semantics.